

Subsurface imaging in shield tunnelling and pipeline localisation via a robust cosmic-ray muography system: a field study

Z. Yao¹, Y. Zhou^{2,*}, L. Huang^{3,*}, S. Liu¹, Y. Wang¹, Z. Zhang¹, Y. Wang², H. Fei², B. Yu⁴, J. Lei³, T. Song³, Z. Zhao³, Z. Xiong³, K. Li³, & Z. Zhang⁵

¹University of Science and Technology of China, Hefei, China

²Jianwei Scientific Instruments (Anhui) Technology Co., Ltd., Hefei, China

³Shenzhen Metro Group Co., Ltd., Shenzhen, China

⁴Jianwei Geophysical Exploration (Shenzhen) Technology Co., Ltd., Shenzhen, China

⁵China Railway No.8 Engineering Group Co., Ltd., Chengdu, China

*Corresponding authors: Yong Zhou (zhouyong@jwky.com);

Liping Huang (huanglp@shenzhenmc.com).

ABSTRACT: Uncertain ground conditions and buried utilities complicate shield tunnelling, while boreholes and surface investigations may provide incomplete coverage in dense urban settings. Cosmic-ray muography offers a passive means of investigating subsurface density, but field deployment requires instrumentation that can operate under construction vibration, elevated temperature, high humidity and restricted access. This paper presents a field-deployable three-layer Micromegas muography system and two applications on Shenzhen Metro Line 22. The system incorporates environmental control, pneumatic vibration isolation and integrated data acquisition for both mobile and stationary measurements. During dynamic deployment aboard tunnel boring machines, it operated at depths of approximately 15–30 m over 1371 rings. On the right line, normal acquisition totalled 2965.5 h, corresponding to 85.2% of the monitoring period and 97.2% after external power and network interruptions were excluded. Measurements at two shield positions were combined with tunnel geometry and borehole stratigraphy to reconstruct density corrections relative to a geological reference model, indicating a possible low-density region above and to one side of the tunnel. In a separate static investigation, observations from four detector positions were processed using position-specific U-Net models and multi-view geometry to estimate a candidate buried-pipeline axis. The result was supplied as a spatial constraint for construction planning, but was not independently confirmed. The deployments demonstrate the field feasibility and decision-support potential of muography for subsurface investigation during tunnel construction, while highlighting the need for matched reference measurements, uncertainty quantification and independent validation.

Keywords: muography; shield tunnelling; density reconstruction; pipeline localisation

1 INTRODUCTION

Uncertain ground conditions and buried utilities can interrupt shield advance and threaten surrounding infrastructure. Boreholes provide direct but spatially discrete geological information, while urban access restrictions and ongoing construction can constrain surface investigations. Measurements acquired from within a tunnel can therefore complement conventional surveys by providing directional information about the ground surrounding the excavation.

Cosmic-ray muography exploits atmospheric muons that retain sufficient energy to reach an underground detector after traversing soil and rock. The transmission approach used here measures the directional flux remaining after passage through the target (Procureur 2018). Particle directions are reconstructed from hit positions in multiple tracking layers, and the underground flux is compared with an open-sky reference to estimate opacity, or density integrated along the muon path. Measurements from different directions and positions can then be combined with tunnel geometry and geological information to investigate density variations. The method is passive and requires neither an artificial radiation source nor direct access to the target.

Previous underground studies have reconstructed overburden density and rock–air interfaces from multiple detector locations (Guardincerri et al. 2017), identified concealed construction

shafts and a subsequently confirmed overburden void in a railway tunnel (Thompson et al. 2020), and detected structural changes along a subway station and tunnel (Mao et al. 2023). These studies establish the potential of tunnel-based measurements, but sustained deployment during active shield construction introduces additional instrumentation and operational requirements.

In particular, mounting a muography system on an operating tunnel boring machine (TBM) exposes the detector and electronics to mechanical vibration, elevated temperature and high humidity. Confined installation space and restricted maintenance access further influence the system design. As the shield advances, the detector position changes relative to the surrounding ground, so each measurement must be associated with its observation geometry. A stationary investigation within a tunnel poses a different requirement: the instrument must be repositioned accurately to combine several views of a fixed target.

This paper presents a field-deployable three-layer Micromegas muography platform supporting both dynamic TBM-mounted measurements and stationary multi-position tunnel investigations. The contributions are: (1) a system combining environmental control, pneumatic isolation and integrated data acquisition for construction conditions; (2) an assessment of sustained operation on Shenzhen Metro Line 22; and (3) two field demonstrations, comprising TBM-mounted ground-density imaging and a four-position investigation of a suspected buried pipeline. Together, these applications examine the instrumentation requirements and decision-support potential of muography during tunnel construction.

2 MUOGRAPHY SYSTEM AND FIELD PERFORMANCE

2.1 *Detector and tracking principle*

The tracking unit comprises three thermally bonded Micromegas layers, each with a $40 \times 40 \text{ cm}^2$ sensitive area and separated by approximately 15 cm (Feng et al. 2021). Orthogonal strip readouts provide two-dimensional hit positions with an effective pitch of 1.6 mm, allowing each muon trajectory to be reconstructed by fitting hits across the three layers. Compact front-end cards digitise the signals, while a central data acquisition board forms inter-layer coincidences and aggregates the events. Optical fibres connect the front-end electronics to the acquisition board, and data are transferred to the control computer via USB 3.0 (Wang et al. 2025; Liu et al. 2025).

2.2 *Field-compatible system design*

The main enclosure integrates the detectors, readout electronics, power supplies and active temperature and humidity controls in a compact arrangement. A separate enclosure houses the argon–carbon-dioxide gas supply, simplifying access to the consumable subsystem. For TBM-mounted operation, four pneumatic isolators separate the inclined detector frame from the shield support platform. These measures were intended to limit environmental and mechanical disturbances while allowing the same transportable instrument to be repositioned for stationary measurements. Figure 1 shows the enclosure design with dimensions (a), internal arrangement (b) and installation on the TBM (c).

2.3 *Field deployment and operational performance*

The system operated at depths of approximately 15–30 m in the Guihua Station–Songyuanxia Station interval from 6 November 2025 to 27 July 2026, covering 1371 rings (428 on the left line and 943 on the right). It was installed approximately 5 m behind the cutterhead on the left line and 15 m behind it on the right. Detector positions and orientations were associated with the corresponding ring records for subsequent analysis. Monitored enclosure temperatures remained below 40 °C and relative humidity below 70%.

On the right line, normal acquisition totalled 2965.5 h, or 85.2% of the 3480.0 h monitoring record. The availability was 97.2% after 401.5 h of external power loss and 28.3 h of network interruption were excluded. The remaining 84.7 h comprised system maintenance. These figures describe the achieved field operation; the available records do not isolate the individual effects of environmental control and pneumatic isolation.

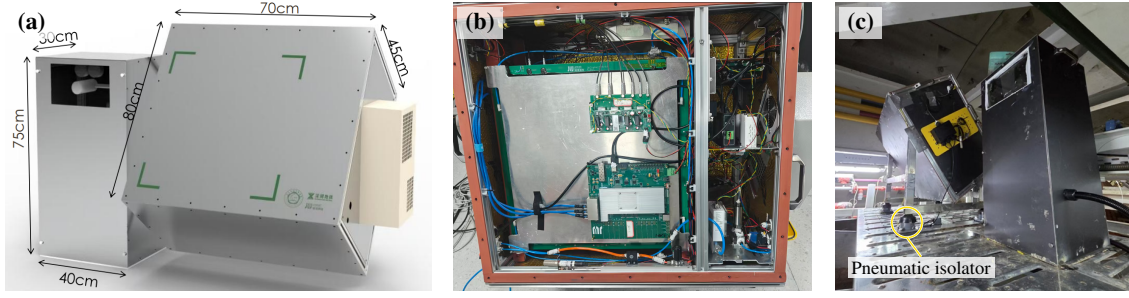


Figure 1. Muography system: (a) enclosure and exterior design with dimensions; (b) photograph showing the internal structure; (c) deployment on the tunnel boring machine.

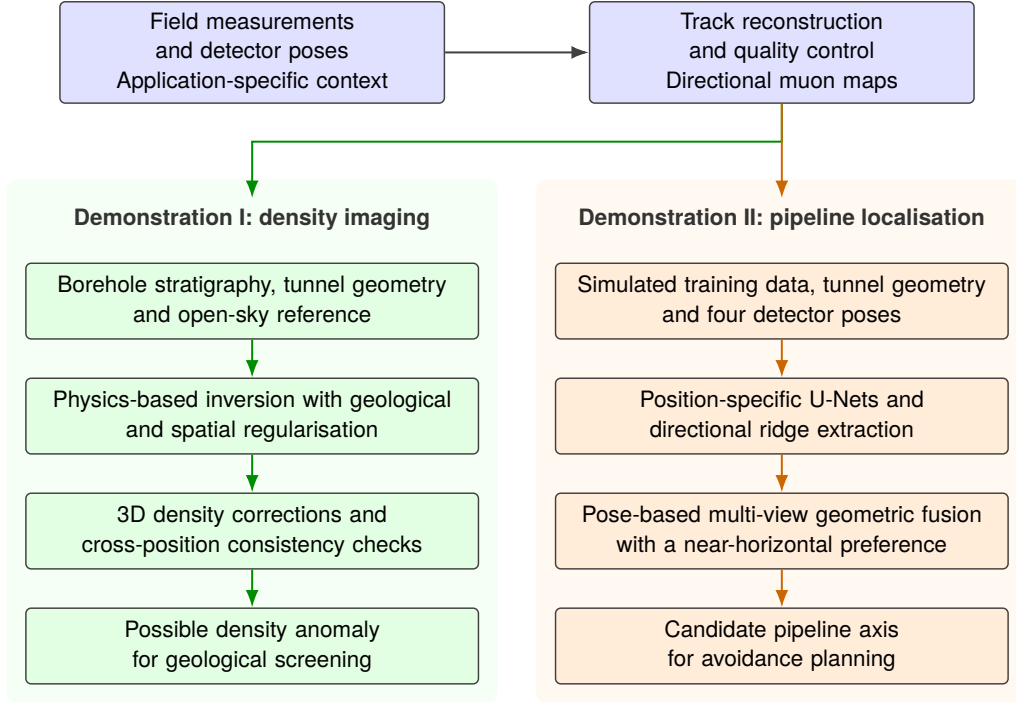


Figure 2. Overall analysis workflow for the two field demonstrations. Common track reconstruction and directional mapping are followed by a physics-based density inversion in Demonstration I and a learning-assisted multi-view localisation procedure in Demonstration II.

The two field demonstrations shared the same measurement chain but used different interpretation strategies, as summarised in Figure 2. The first used physics-based inversion to estimate density corrections relative to a geological model, whereas the second combined learned directional-feature extraction with multi-view geometry to estimate a candidate pipeline axis.

3 FIELD DEMONSTRATION I: TBM-MOUNTED GROUND-DENSITY IMAGING

3.1 Engineering objective and site conditions

The first demonstration examined whether measurements acquired during shield advance could identify density variations relative to the available geological model. Near the Fuzhou Avenue bridge, borehole stratigraphy provided a reference and two successive shield positions provided different viewing geometries. The result was intended for geological screening rather than confirmation of a specific feature.

3.2 Measurements and reference model

The primary dataset contains 5240 and 6840 selected underground tracks at rings 1672 and 1679, respectively. Underground exposures were 21.37 and 34.00 h. The corresponding open-sky reference measurements contained 457,255 selected tracks, with exposures of 20.97 and 66.52 h. The same detector hardware, track-reconstruction procedure and selection criteria were used for the open-sky and underground measurements.

In the geological survey frame, Y is upward, Z points towards decreasing ring number and X is transverse. Ring 1672 defines the longitudinal origin and vertical track reference. The detector centres at rings 1672 and 1679 are represented by $(2, 0, 0)$ and $(2, 0.27, -12.66)$ m, respectively. The initial detector-normal elevation was 33.2° , varying within $\pm 0.6^\circ$ during advance.

The geological model uses 11 borehole control points. Table 1 gives the layer-top depths and assigned reference densities at five boreholes around the measurement interval. Boundaries are interpolated along the tunnel and extended uniformly across X . The tunnel inner and outer diameters are 8.0 and 8.8 m; the cavity and 0.4 m lining are modelled separately, with a lining density of 2.4 g cm^{-3} .

Table 1. Borehole locations and layer-top depths below ground (m), with assigned densities (g cm^{-3}).

Density	TJL-39 (-0.87, 62.05)	TJL-38 (8.32, 17.19)	TJL-37 (10.02, -17.19)	THS-2-1 (8.86, -43.57)	THS-2 (1.38, -73.94)
1.95	0.00	0.00	0.00	0.00	0.00
2.20	3.00	0.00	5.00	6.60	6.00
1.85	3.00	8.50	5.00	6.60	6.00
1.95	20.00	18.90	18.50	18.00	17.60
2.20	22.50	25.40	24.80	26.00	20.00
2.25	23.70	34.30	42.30	31.00	22.00
2.40	23.70	34.30	44.50	31.00	22.90
2.45	26.50	37.80	–	33.80	27.40

Coordinates in parentheses are (X, Z) in metres. All borehole identifiers have the prefix M33Z3-. Layers are ordered from shallow to deep; equal successive depths indicate zero thickness.

3.3 Algorithm design

The reconstruction used a physics-based comparison between measured and predicted directional counts. Its inputs were the underground observations from the two shield positions, the open-sky reference, detector poses, tunnel geometry and borehole-derived geological model. For each trial density distribution on a 2 m grid, ray tracing calculated the material thickness traversed in each direction (Siddon 1985). The resulting opacities were converted to expected muon transmission using the spectrum of Guan et al. (2015) and standard-rock energy-loss ranges (Groom et al. 2001). A Poisson count model then compared the predictions with the sky and underground measurements. A common factor of 0.53 normalised the two underground positions; it was estimated from background directions excluding known bridge piles and may also absorb differences between the sky and underground datasets.

Because only two viewing positions were available, the inversion was constrained by the geological reference and by spatial continuity. Departures from the borehole model were penalised more strongly near boreholes, while smoothing between neighbouring grid cells was reduced across interpreted layer boundaries. The open-sky directional rates and density corrections were estimated jointly within physical bounds. The reconstructed volume therefore represents regularised corrections relative to the borehole model, rather than an independent absolute-density estimate.

The analysis covered $X = [-48, 30]$, $Y = [-6, 36]$ and $Z = [-24, 42]$ m and fitted 151 adaptive angular bins with widths of 6° or 12° . The two positions were reconstructed jointly to obtain the main result and separately to examine consistency. Each single-position reconstruction was also used to predict the counts at the other position, providing a cross-position check based on Poisson deviance.

3.4 Results and consistency checks

Figure 3 compares count-derived effective opacity with the borehole reference. Both positions show negative differences at large zenith angles, whereas positive differences are more prominent at ring 1679. Negative values indicate less integrated density than predicted by the reference model.

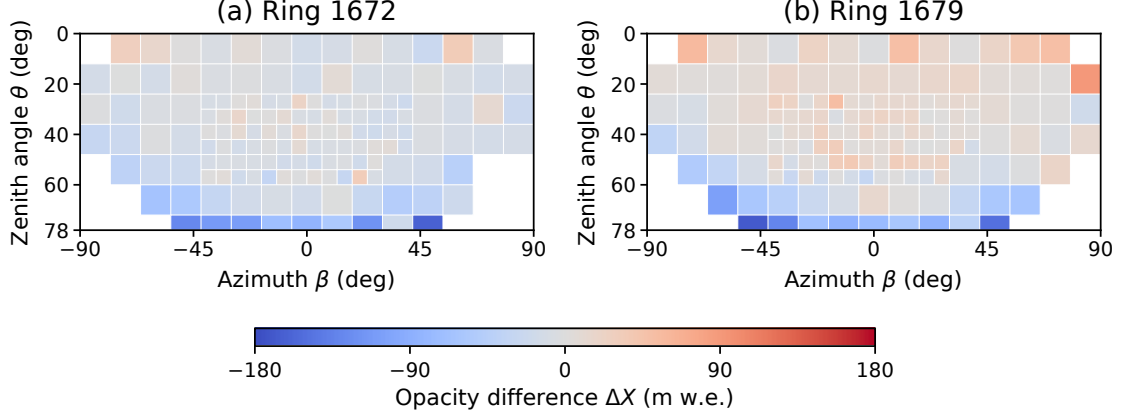


Figure 3. Effective opacity differences, $\Delta X = X_{\text{obs}} - X_{\text{reference}}$, for (a) ring 1672 and (b) ring 1679.

Combining the two viewing geometries gives the density-correction distribution in Figure 4. A possible low-density region appears above and to the negative- X side of the tunnel. A positive correction near ring 1679 reaches $+0.57 \text{ g cm}^{-3}$ at $(1, 9, -9) \text{ m}$, a location directly sampled by that position.

A fixed set of voxels above the tunnel on its negative- X side was selected from a preliminary joint reconstruction and retained for all fits. Its joint mean correction is -0.36 g cm^{-3} ; separate fits to rings 1672 and 1679 give -0.21 and -0.24 g cm^{-3} , although their spatial distributions differ.

Cross-position predictions provide a further consistency check using Poisson deviance, where smaller values indicate closer agreement. Predicting ring 1679 from the ring-1672 reconstruction reduces the deviance from 801.4 to 770.6; the reverse prediction reduces it from 272.1 to 256.9. The reductions are modest but directionally consistent with the negative regional means.

3.5 Interpretation and limitations

The result indicates a possible low-density region relative to the borehole model, not a confirmed void or material. Interpretation is limited by sparse borehole control, interpolated layer boundaries, non-simultaneous sky references, normalisation and regularisation. Because the comparison region was selected from a preliminary reconstruction, its mean is descriptive rather than an independent test statistic.

4 FIELD DEMONSTRATION II: STATIC MULTI-POSITION PIPELINE INVESTIGATION

4.1 Engineering objective and survey arrangement

Conventional detection methods had not located the pipeline at the position indicated in the original drawings, leaving uncertainty about its location relative to the planned tunnel alignment. The muography survey was therefore undertaken to investigate whether additional positional information could be obtained to inform pipeline-avoidance measures during shield advance.

The detector was deployed sequentially at four stationary positions within the tunnel, as shown in Figure 5. Acquisition times at P0–P3 were 22.8, 21.7, 32.5 and 22.0 h, respectively. Tracks were binned into 36×38 directional maps, with 21,084–31,309 underground counts per position and 78,718 counts in the shared open-sky map.

The coordinate origin for this survey is the centre of ring 252; X is along the tunnel opposite to advance, Y points horizontally towards the reference wall and Z is upward. Table 2 gives the instrument centres and orientations used in the reconstruction. Pitch is the elevation of the detector

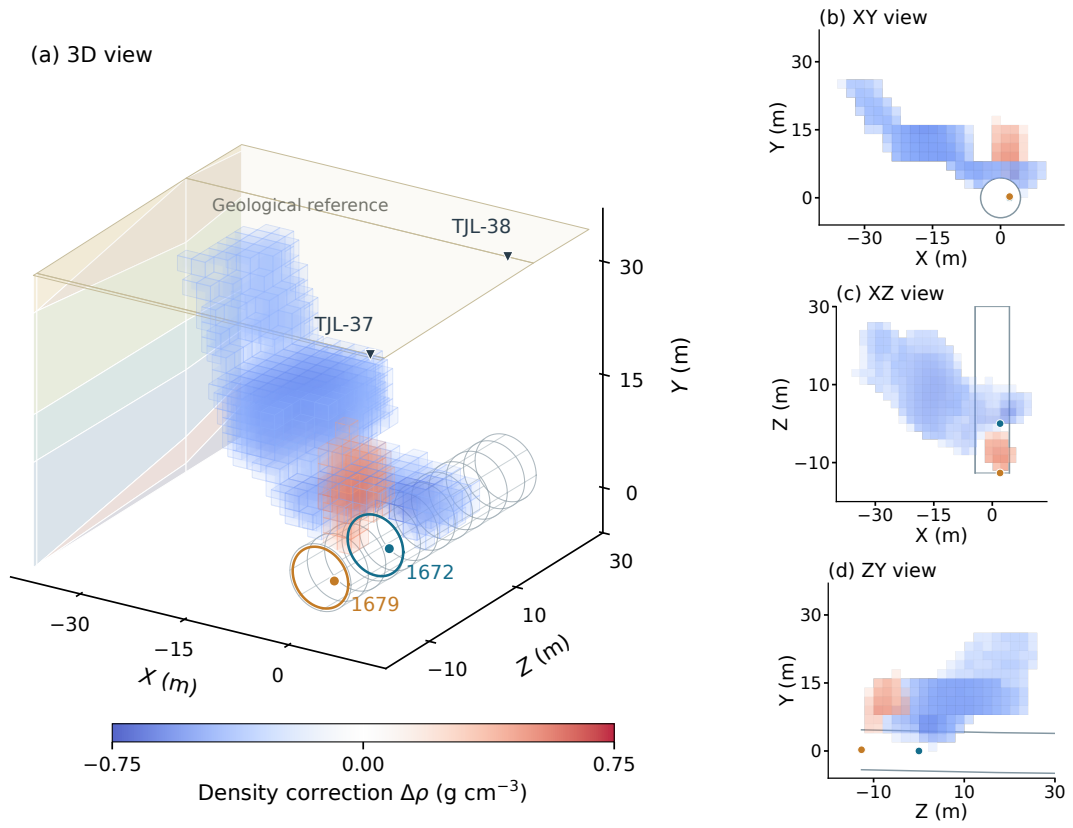


Figure 4. Joint density corrections relative to the borehole model: (a) three-dimensional rendering and (b–d) XY, XZ and ZY views. Only 2 m voxels with direct ray coverage and $|\Delta\rho| \geq 0.35 \text{ g cm}^{-3}$ are shown; transparency decreases with $|\Delta\rho|$.

normal; $\alpha = 0$ faces positive Y , with positive rotation towards negative X . The geometric model follows the construction drawings, with tunnel inner/outer radii of 2.75/3.10 m and ground at $Z = 16.8 \text{ m}$.

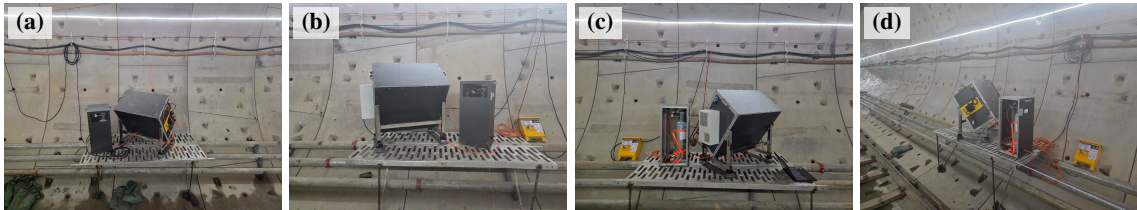


Figure 5. Field deployment for the pipeline survey at (a) P0, (b) P1, (c) P2 and (d) P3.

Table 2. Pipeline detector centres and orientations in the ring-252 frame.

Position	$X \text{ (m)}$	$Y \text{ (m)}$	$Z \text{ (m)}$	Pitch ($^\circ$)	$\alpha \text{ (}^\circ\text{)}$
P0	-0.05	1.38	-1.50	36.3	-32.4
P1	3.75	1.38	-1.50	34.3	17.1
P2	7.50	1.48	-1.50	34.9	35.0
P3	-3.80	1.60	-1.50	35.8	-77.0

4.2 Algorithm design

The pipeline workflow separated directional-feature detection from three-dimensional localisation. First, a position-specific U-Net (Ronneberger et al. 2015) was trained on simulated directional maps containing pipelines with varied locations and orientations, together with cases containing no pipeline. Each fixed network then converted its field observation into a relative feature-score map. These scores identify pipeline-like directional structure but are not calibrated probabilities of pipeline presence. Evaluation on 50,000 simulated test samples per position gave pixel ROC-AUC values of 0.9585–0.9703. As shown in Figure 6, all four field score maps contained an extended ridge.

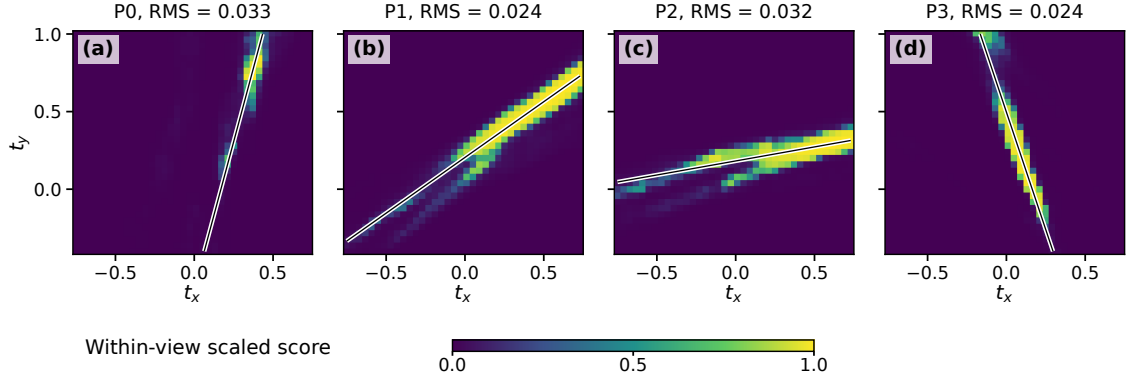


Figure 6. Pipeline directional maps at (a) P0, (b) P1, (c) P2 and (d) P3. Colours represent within-view scaled network scores; lines mark fitted ridges. Coordinates are detector-local slopes; RMS denotes the weighted root-mean-square residual transverse to each ridge.

The localisation stage fitted a weighted straight ridge to the high-score region in each directional map. The measured detector poses transformed these view-dependent ridges into backprojection planes in the common tunnel coordinate system. The candidate pipeline axis was found as the line most consistent with all four planes, with empirical view weights based on ridge linearity and transverse residual. A unit-weight soft preference for a near-horizontal direction incorporated the expected utility geometry without enforcing exact horizontality. This multi-view stage converted the four directional features into a single spatial constraint for construction planning.

4.3 Candidate axis and engineering use

The fitted candidate axis is

$$\mathbf{x}(\ell) \simeq [-3.41, 0.38, 6.50]^T \text{ m} + \ell[0.69, 0.73, -0.05]^T, \quad (1)$$

where ℓ is distance along the axis in metres. As shown in Figure 7, the candidate passes above the tunnel near ring 252.

The candidate position and shield-avoidance recommendations were supplied to the construction team as precautionary spatial information. The team reported implementing avoidance measures based on this information.

4.4 Limitations

The line is a candidate axis, not independent confirmation. The networks were evaluated on simulated data and their field outputs are not calibrated probabilities. The axis also depends on detector poses, empirical weights and the horizontal penalty; no positional uncertainty or independent confirmation was available.

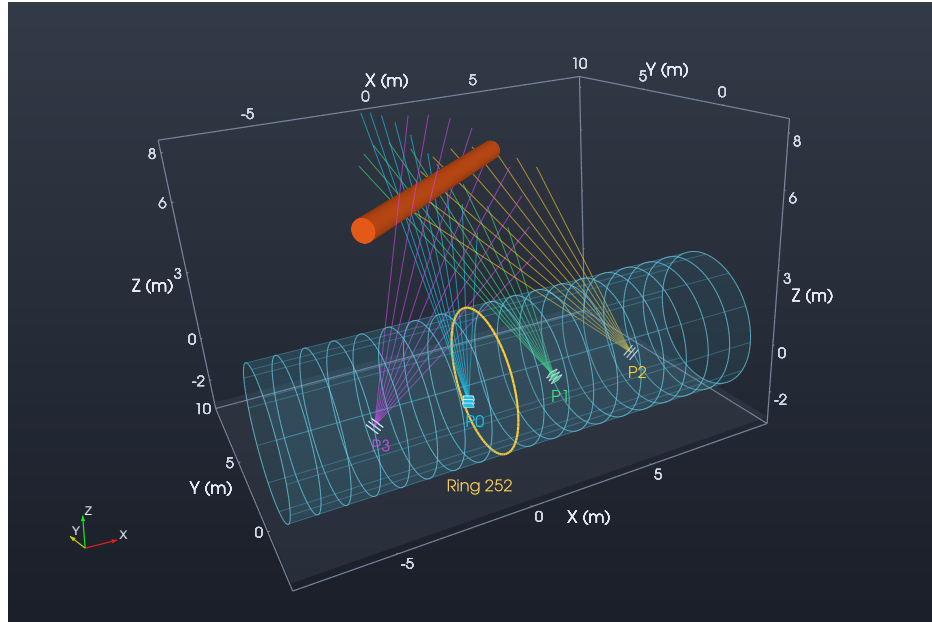


Figure 7. Three-dimensional pipeline candidate axis with the horizontal soft constraint. Coloured ray fans represent the backprojection planes from P0–P3; the orange tube shows the candidate pipe with its reported diameter of 0.8 m.

5 DISCUSSION

5.1 Instrumentation lessons

The campaign shows that one transportable Micromegas system can support dynamic TBM-mounted and stationary multi-position measurements. Most right-line downtime arose from external power and network interruptions, making site infrastructure part of the deployment design.

5.2 Investigation capability

Acquisition required approximately 21–34 h per position, making the present system more suitable for targeted screening than immediate warning. Directional measurements should be combined with boreholes, construction geometry and conventional investigations rather than treated as stand-alone confirmation.

5.3 Current limitations and development needs

Quantitative interpretation remains limited by reference compatibility, sparse viewpoints and unquantified uncertainty. Future work should use matched sky references, systematic pose records, additional views, sensitivity studies and independently confirmed targets.

6 CONCLUSIONS

A three-layer Micromegas system completed sustained field deployment on Shenzhen Metro Line 22. Right-line acquisition availability was 85.2% overall and 97.2% after external interruptions were excluded. Dynamic measurements indicated a possible low-density region relative to the borehole model, with a mean correction of -0.36 g cm^{-3} in the selected region. A static four-position investigation produced a candidate pipeline axis for avoidance planning. These results establish field feasibility and decision-support potential, but require improved reference measurements, additional viewpoints, uncertainty quantification and independent validation.

ACKNOWLEDGEMENTS

The authors acknowledge Shenzhen Metro Group Co., Ltd. for field access and coordination, the University of Science and Technology of China for research support, the Jianwei teams for equipment and maintenance, and the site personnel for assistance with installation and data acquisition.

REFERENCES

- Feng, J., Zhang, Z., Liu, J., Qi, B., Wang, A., Shao, M. & Zhou, Y. 2021. A thermal bonding method for manufacturing Micromegas detectors. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 989: 164958.
- Groom, D. E., Mokhov, N. V. & Striganov, S. I. 2001. Muon stopping power and range tables: 10 MeV–100 TeV. *Atomic Data and Nuclear Data Tables* 78(2): 183–356.
- Guan, M., Chu, M.-C., Cao, J., Luk, K.-B. & Yang, C. 2015. A parametrization of the cosmic-ray muon flux at sea-level. arXiv:1509.06176.
- Guardincerri, E., Rowe, C., Schultz-Fellenz, E., Roy, M., George, N., Morris, C., Bacon, J., Durham, M., Morley, D., Plaud-Ramos, K., Poulson, D., Baker, D., Bonneville, A. & Kouzes, R. 2017. 3D cosmic ray muon tomography from an underground tunnel. *Pure and Applied Geophysics* 174(5): 2133–2141.
- Liu, J., Wang, Y., Feng, C., Liu, S. & Chen, Q. 2025. A back-end electronics based on optical fiber communication for small to medium-scale physics experiments. *Journal of Instrumentation* 20(05): T05002.
- Mao, X., Li, Z., Dong, S., Li, J., Zhang, J., Pang, J., Cheng, Y., Liao, B., Ouyang, X. & Han, R. 2023. Muon radiography experiments on the subway overburden structure detection. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 1055: 168391.
- Procureur, S. 2018. Muon imaging: Principles, technologies and applications. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 878: 169–179.
- Ronneberger, O., Fischer, P. & Brox, T. 2015. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*. Vol. 9351. Lecture Notes in Computer Science. Cham: Springer, pp. 234–241.
- Siddon, R. L. 1985. Fast calculation of the exact radiological path for a three-dimensional CT array. *Medical Physics* 12(2): 252–255.
- Thompson, L. F., Stowell, J. P., Fargher, S. J., Steer, C. A., Loughney, K. L., O’Sullivan, E. M., Gluyas, J. G., Blaney, S. W. & Pidcock, R. J. 2020. Muon tomography for railway tunnel imaging. *Physical Review Research* 2(2): 023017.
- Wang, T., Wang, Y., Yao, Z., Liu, Y., Feng, C., Zhang, Z. & Liu, S. 2025. A compact readout electronics based on current amplifier for Micromegas detector in muon imaging. *IEEE Transactions on Nuclear Science* 72(3): 414–420.